golden_teacher

Building manual – Legolize it



Parts	220, 90 shaped, 21 special
Layers	20 (bricks, 9.6 mm)
Size	136 × 136 × 192 mm
Grid	17 × 17 × 20 cells, pitch 8 mm
Stud play	0.15 mm (wall 1.35, tube 6.2137, socket 5.1 mm)
Holds together	no: 5 groups, 4 floating

Parts

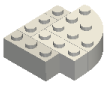
Part	Colour	Count
2x6	white	1
1x8	tan	1
1x8	white	2
2x4	black	1
2x4	bright light orange	3
1x6	black	1
1x6	light yellow	1
1x6	tan	2
1x6	white	1
2x3	black	3
2x3	light yellow	1
2x3	tan	2
2x3	white	3
1x4	black	3
1x4	dark grey	1
1x4	pearl gold	1
1x4	tan	1
2x2	black	1
2x2	tan	1
2x2	white	2
1x3	black	4
1x3	bright light orange	1
1x3	light yellow	1
1x3	tan	1
1x3	white	7
1x2	black	5
1x2	bright light orange	3
1x2	cream	3
1x2	dark grey	3
1x2	light yellow	4
1x2	tan	1
1x2	white	13
1x1	black	7
1x1	bright light orange	3
1x1	cream	2
1x1	dark grey	2
1x1	light yellow	2
1x1	pearl gold	1
1x1	tan	2
1x1	white	12
round corner 3x3	white	2
slope 3x2	bright light orange	8
slope 3x2	dark orange	1
slope 3x2	light yellow	1
dome corner 2x2	bright light orange	4
dome corner 2x2	light yellow	2
inverted curve 2x2	black	12
round corner 2x2	white	5
slope 2x2	black	4
slope 2x2	bright light orange	1
slope 2x2	light yellow	4

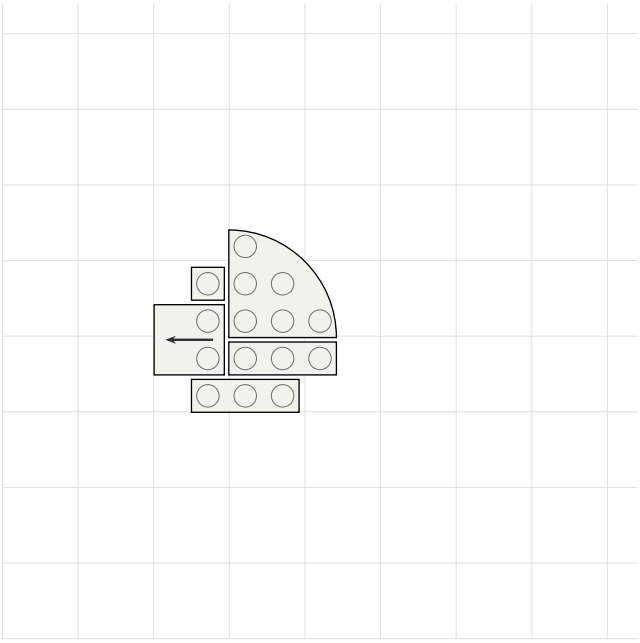
slope 2x2	white	1
inverted curve 3x1	black	1
inverted slope 3x1	black	12
inverted slope 3x1	dark grey	2
slope 3x1	bright light orange	3
slope 3x1	dark orange	1
slope 3x1	light yellow	2
slope 3x1	pearl gold	1
inverted slope 2x1	white	2
slope 2x1	dark grey	1
slope 2x1	light yellow	4
slope 2x1	white	1
special 1x2	black	1
special 1x2	dark orange	1
special 1x2	white	2
dome corner 1x1	white	4
round corner 1x1	white	8
slope 1x1	dark orange	1
slope 1x1	light yellow	1
slope 1x1	pearl gold	1
special 1x1	black	9
special 1x1	bright light orange	1
special 1x1	light yellow	1
special 1x1	pearl gold	1
special 1x1	tan	1
special 1x1	white	4
all		220

How to read the steps

Each step is one layer. The picture shows the model so far, the new layer bright. The plan shows the layer from above: the layer below in grey, every part in its colour, circles for studs. An arrow points down the fall of a slope or curved slope (dashed: an inverted one, falling underneath). Quarter circles are round or dome corners, full circles round bricks. Hatched parts marked **S***n* are special pieces: custom parts with the input's exact shape, printed from the STL stack (`bricks/special_NNNNN.stl`).

Step 1 (layer 0, z = 0 mm, 5 parts)

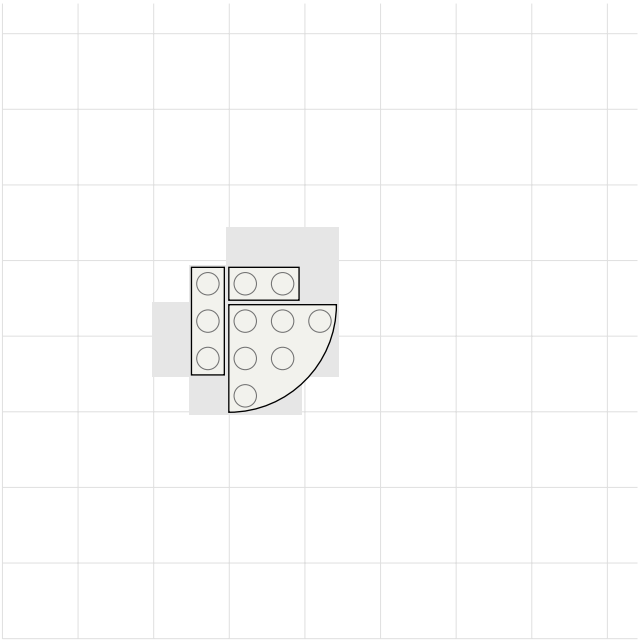




Part	Colour	Count
1x3	white	2
1x1	white	1
round corner 3x3	white	1
slope 2x2	white	1

Step 2 (layer 1, z = 9.6 mm, 3 parts)

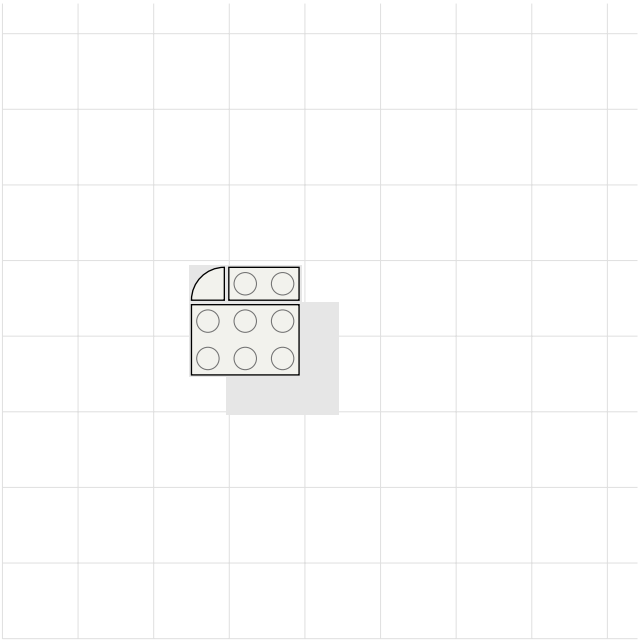




Part	Colour	Count
1x3	white	1
1x2	white	1
round corner 3x3	white	1

Step 3 (layer 2, z = 19.2 mm, 3 parts)

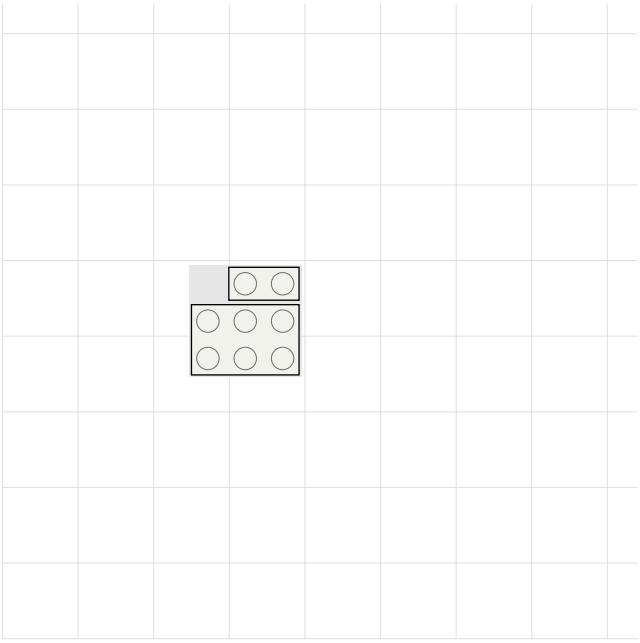




Part	Colour	Count
2x3	white	1
1x2	white	1
dome corner 1x1	white	1

Step 4 (layer 3, z = 28.8 mm, 2 parts)

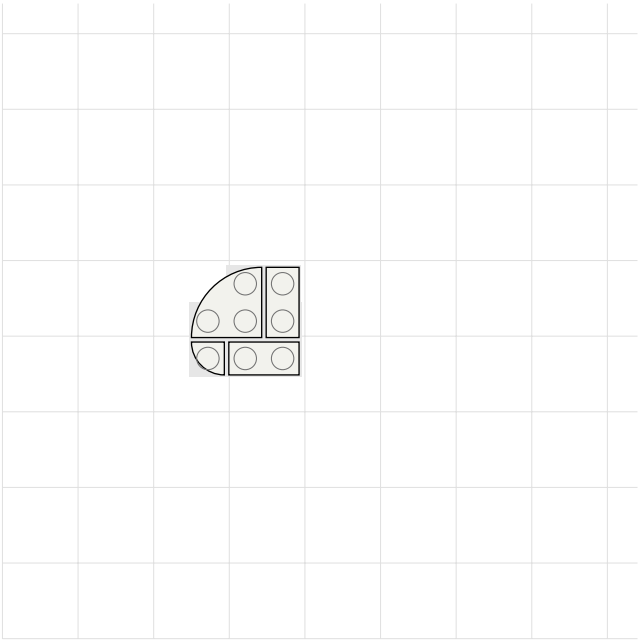




Part	Colour	Count
2x3	white	1
1x2	white	1

Step 5 (layer 4, z = 38.4 mm, 4 parts)

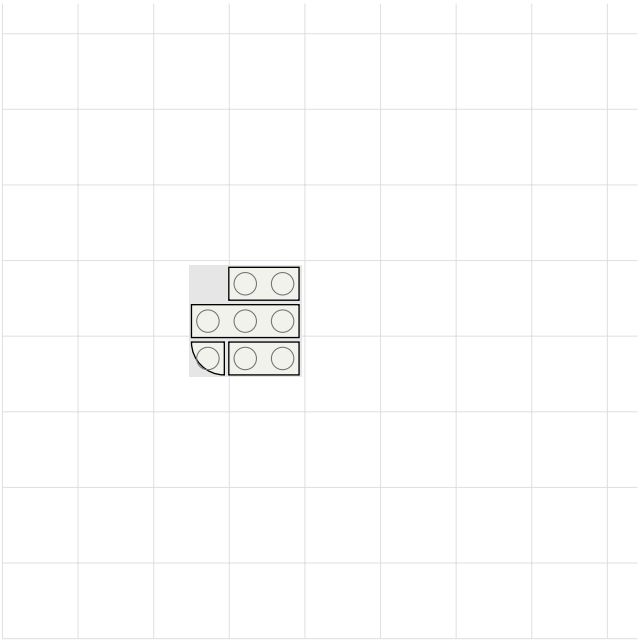




Part	Colour	Count
1x2	white	2
round corner 2x2	white	1
round corner 1x1	white	1

Step 6 (layer 5, z = 48 mm, 4 parts)

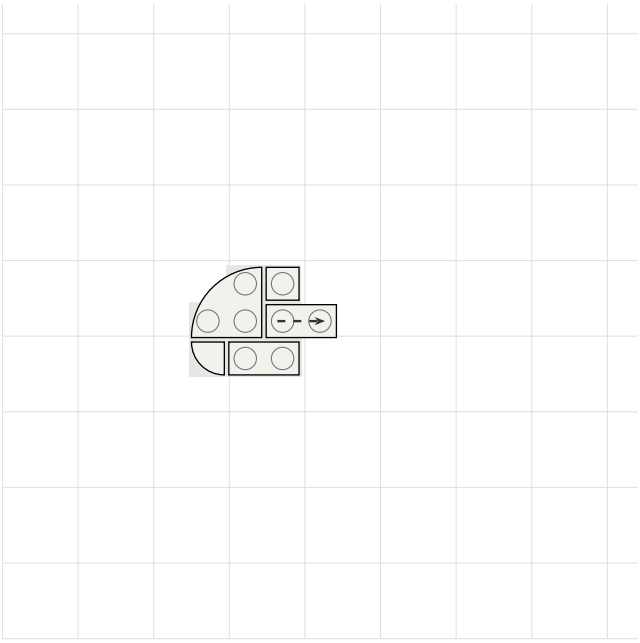




Part	Colour	Count
1x3	white	1
1x2	white	2
round corner 1x1	white	1

Step 7 (layer 6, z = 57.6 mm, 5 parts)

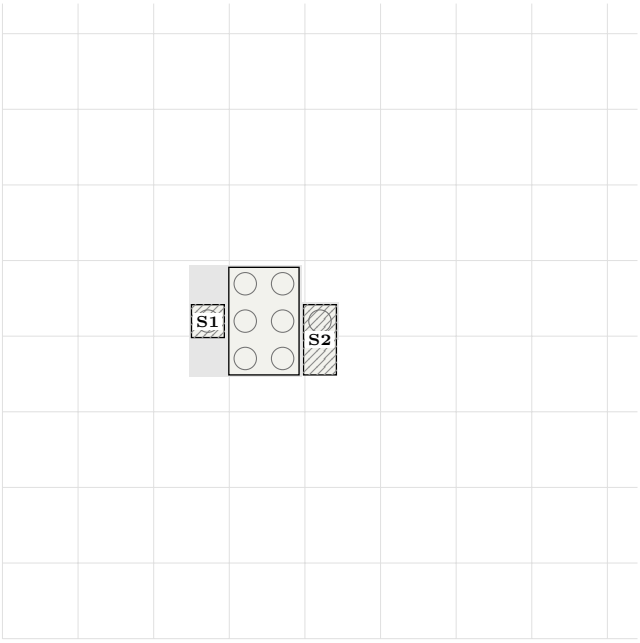




Part	Colour	Count
1x2	white	1
1x1	white	1
round corner 2x2	white	1
inverted slope 2x1	white	1
dome corner 1x1	white	1

Step 8 (layer 7, z = 67.2 mm, 3 parts)

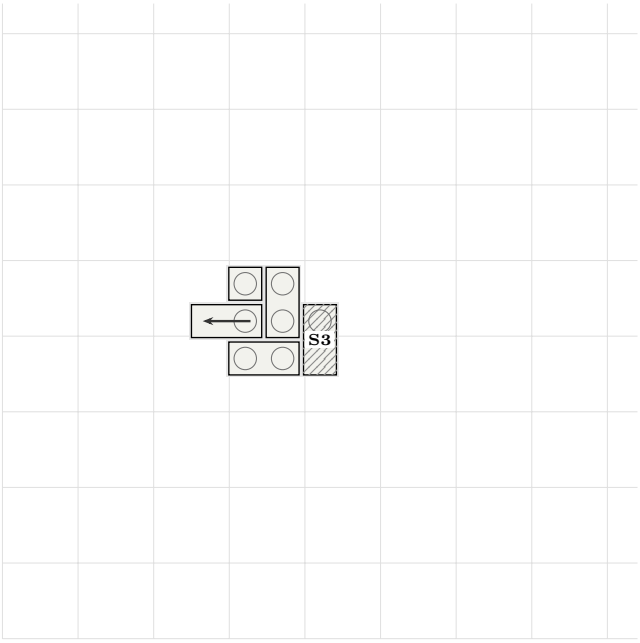




Part	Colour	Count
2x3	white	1
special 1x2	white	1
special 1x1	white	1

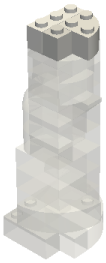
Step 9 (layer 8, z = 76.8 mm, 5 parts)

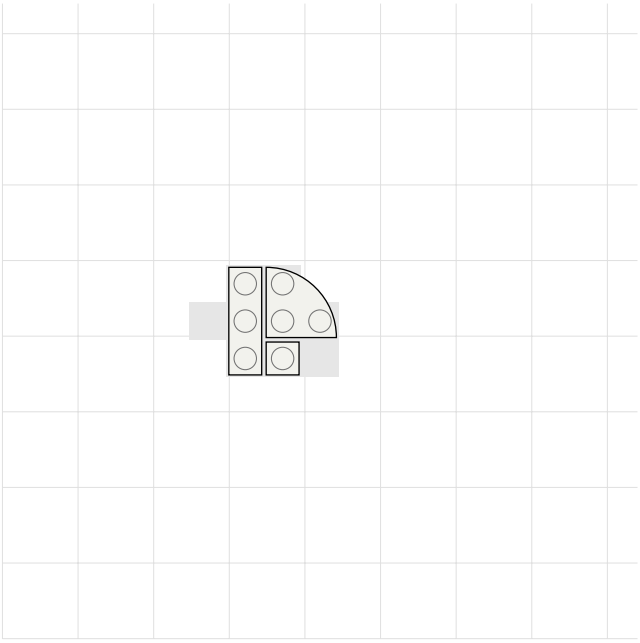




Part	Colour	Count
1x2	white	2
1x1	white	1
slope 2x1	white	1
special 1x2	white	1

Step 10 (layer 9, z = 86.4 mm, 3 parts)

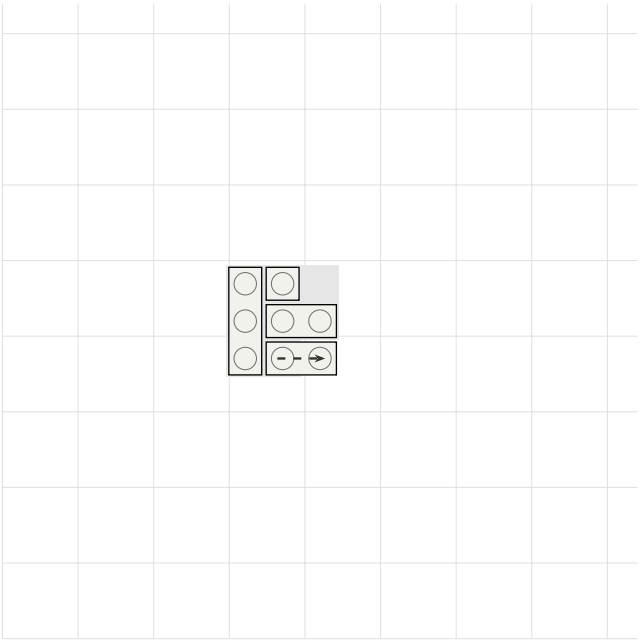




Part	Colour	Count
1x3	white	1
1x1	white	1
round corner 2x2	white	1

Step 11 (layer 10, z = 96 mm, 4 parts)

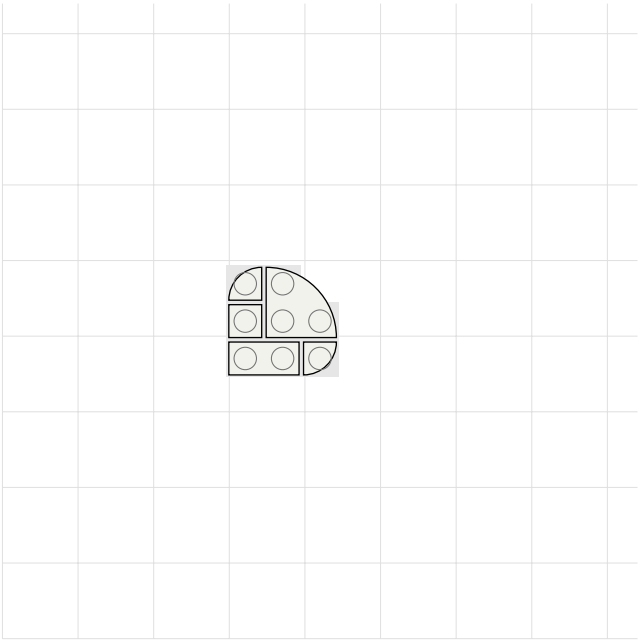




Part	Colour	Count
1x3	white	1
1x2	white	1
1x1	white	1
inverted slope 2x1	white	1

Step 12 (layer 11, z = 105.6 mm, 5 parts)

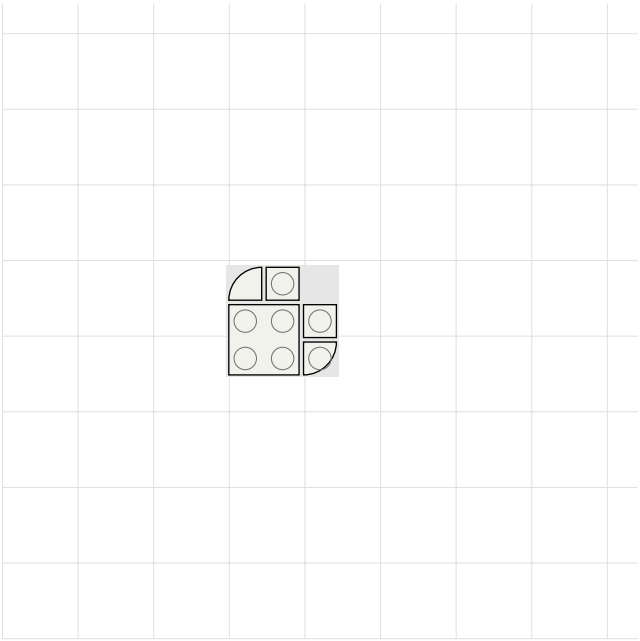




Part	Colour	Count
1x2	white	1
1x1	white	1
round corner 2x2	white	1
round corner 1x1	white	2

Step 13 (layer 12, z = 115.2 mm, 5 parts)

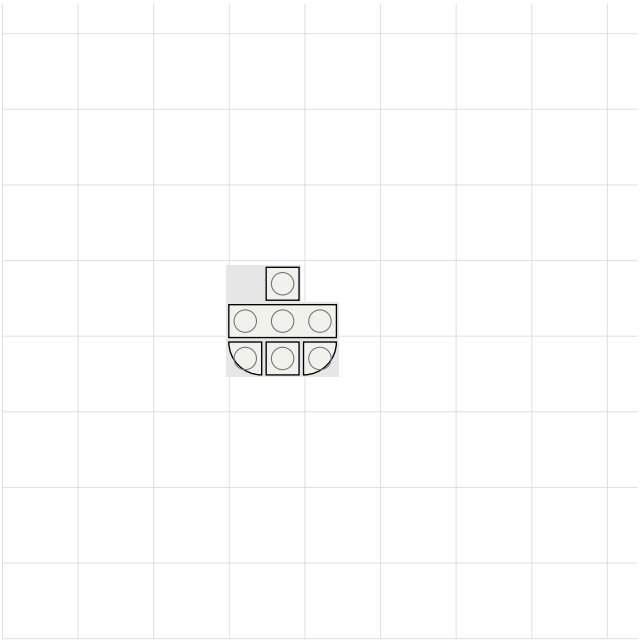




Part	Colour	Count
2x2	white	1
1x1	white	2
dome corner 1x1	white	1
round corner 1x1	white	1

Step 14 (layer 13, z = 124.8 mm, 5 parts)

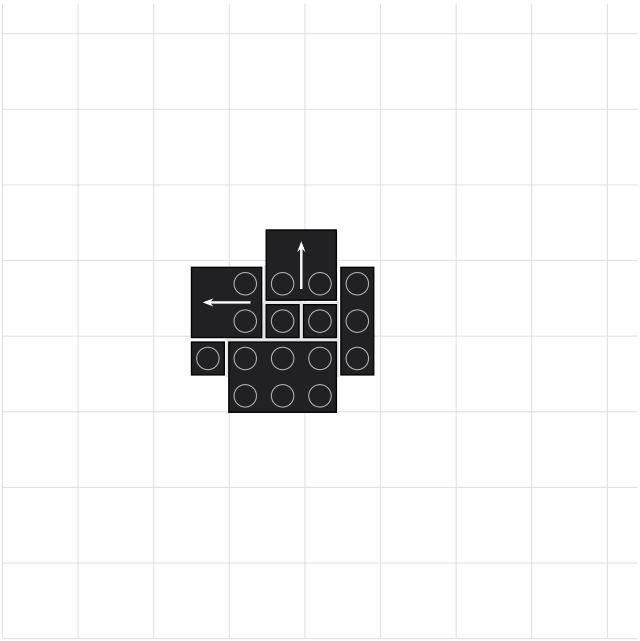




Part	Colour	Count
1x3	white	1
1x1	white	2
round corner 1x1	white	2

Step 15 (layer 14, z = 134.4 mm, 7 parts)

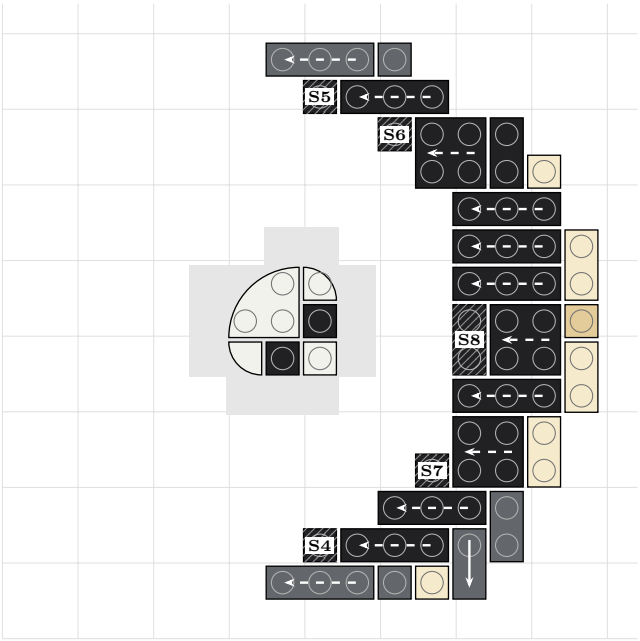




Part	Colour	Count
2x3	black	1
1x3	black	1
1x1	black	3
slope 2x2	black	2

Step 16 (layer 15, z = 144 mm, 34 parts)



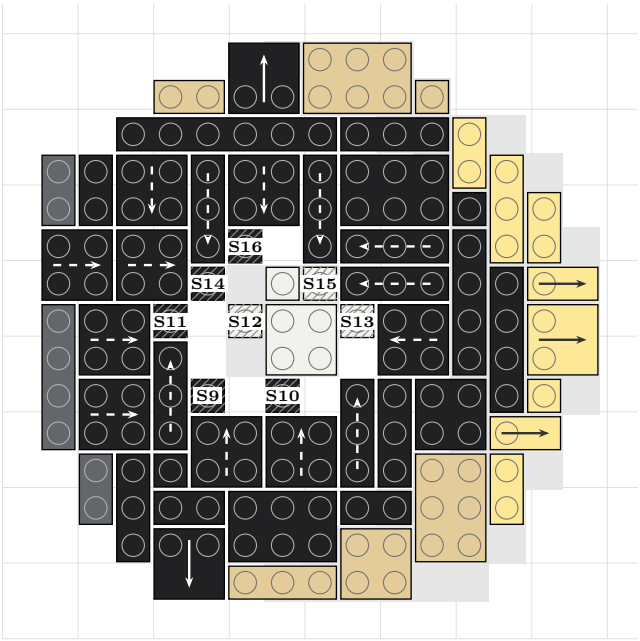


Part	Colour	Count
1x2	black	1
1x2	cream	3
1x2	dark grey	1
1x1	black	2
1x1	cream	2
1x1	dark grey	2
1x1	tan	1
1x1	white	1
inverted curve 2x2	black	3

Part	Colour	Count
round corner 2x2	white	1
inverted curve 3x1	black	1
inverted slope 3x1	black	6
inverted slope 3x1	dark grey	2
slope 2x1	dark grey	1
special 1x2	black	1
dome corner 1x1	white	1
round corner 1x1	white	1
special 1x1	black	4

Step 17 (layer 16, z = 153.6 mm, 58 parts)



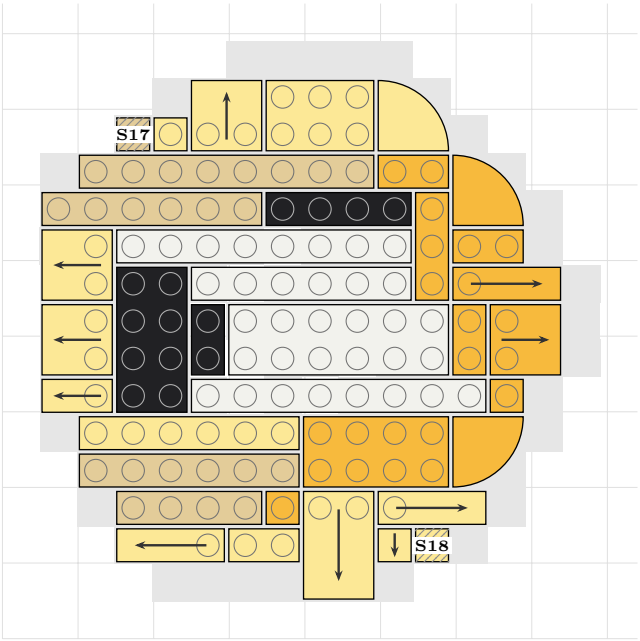


Part	Colour	Count
1x6	black	1
2x3	black	2
2x3	tan	2
1x4	black	2
1x4	dark grey	1
2x2	black	1
2x2	tan	1
2x2	white	1
1x3	black	3
1x3	light yellow	1
1x3	tan	1
1x2	black	3
1x2	dark grey	2

Part	Colour	Count
1x2	light yellow	3
1x2	tan	1
1x1	black	2
1x1	light yellow	1
1x1	tan	1
1x1	white	1
inverted curve 2x2	black	9
slope 2x2	black	2
slope 2x2	light yellow	1
inverted slope 3x1	black	6
slope 2x1	light yellow	2
special 1x1	black	5
special 1x1	white	3

Step 18 (layer 17, z = 163.2 mm, 37 parts)



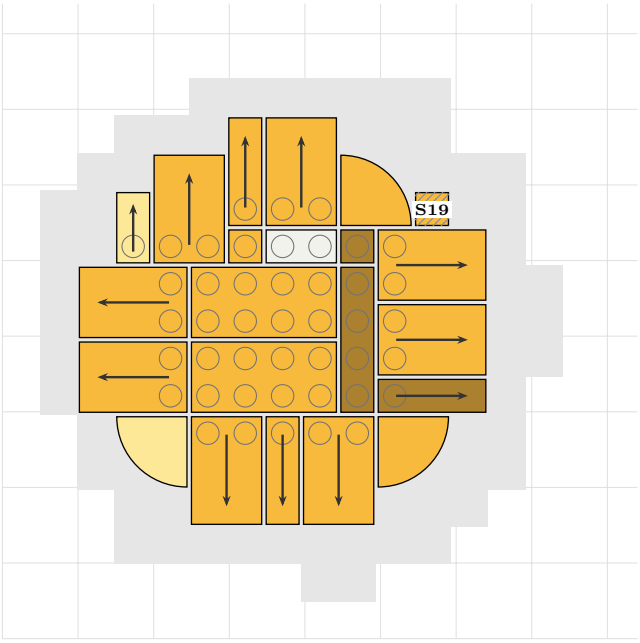


Part	Colour	Count
2x6	white	1
1x8	tan	1
1x8	white	2
2x4	black	1
2x4	bright light orange	1
1x6	light yellow	1
1x6	tan	2
1x6	white	1
2x3	light yellow	1
1x4	black	1
1x4	tan	1
1x3	bright light orange	1
1x2	black	1
1x2	bright light orange	3

Part	Colour	Count
1x2	light yellow	1
1x1	bright light orange	2
1x1	light yellow	1
slope 3x2	light yellow	1
dome corner 2x2	bright light orange	2
dome corner 2x2	light yellow	1
slope 2x2	bright light orange	1
slope 2x2	light yellow	3
slope 3x1	bright light orange	1
slope 3x1	light yellow	2
slope 2x1	light yellow	1
slope 1x1	light yellow	1
special 1x1	light yellow	1
special 1x1	tan	1

Step 19 (layer 18, z = 172.8 mm, 22 parts)

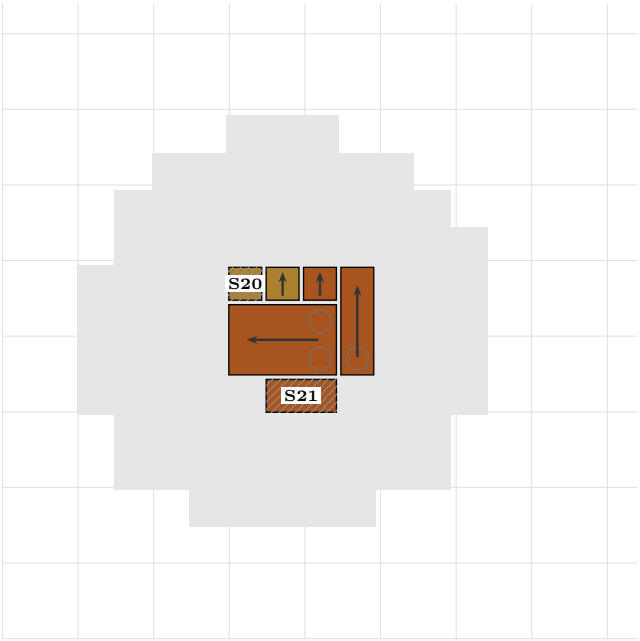




Part	Colour	Count
2x4	bright light orange	2
1x4	pearl gold	1
1x2	white	1
1x1	bright light orange	1
1x1	pearl gold	1
slope 3x2	bright light orange	8
dome corner 2x2	bright light orange	2
dome corner 2x2	light yellow	1
slope 3x1	bright light orange	2
slope 3x1	pearl gold	1
slope 2x1	light yellow	1
special 1x1	bright light orange	1

Step 20 (layer 19, z = 182.4 mm, 6 parts)





Part	Colour	Count
slope 3x2	dark orange	1
slope 3x1	dark orange	1
special 1x2	dark orange	1
slope 1x1	dark orange	1
slope 1x1	pearl gold	1
special 1x1	pearl gold	1

Special pieces

Print each from `bricks/` of the STL stack; the studs and sockets sit where it joins the bricks.

Piece	File	Layer	Cells
S1	`special_00026.stl`	7	1×1 at (5, 8)
S2	`special_00027.stl`	7	1×2 at (8, 7)
S3	`special_00030.stl`	8	1×2 at (8, 7)
S4	`special_00079.stl`	15	1×1 at (8, 2)
S5	`special_00080.stl`	15	1×1 at (8, 14)
S6	`special_00081.stl`	15	1×1 at (10, 13)
S7	`special_00082.stl`	15	1×1 at (11, 4)
S8	`special_00083.stl`	15	1×2 at (12, 7)
S9	`special_00117.stl`	16	1×1 at (5, 6)
S10	`special_00118.stl`	16	1×1 at (7, 6)
S11	`special_00119.stl`	16	1×1 at (4, 8)
S12	`special_00120.stl`	16	1×1 at (6, 8)
S13	`special_00121.stl`	16	1×1 at (9, 8)
S14	`special_00122.stl`	16	1×1 at (5, 9)
S15	`special_00123.stl`	16	1×1 at (8, 9)
S16	`special_00124.stl`	16	1×1 at (6, 10)
S17	`special_00168.stl`	17	1×1 at (3, 13)
S18	`special_00169.stl`	17	1×1 at (11, 2)
S19	`special_00207.stl`	18	1×1 at (11, 11)
S20	`special_00218.stl`	19	1×1 at (6, 9)
S21	`special_00219.stl`	19	2×1 at (7, 6)